Address History Analysis: Eviction Persistence and Mover Trajectories

Philly Evictions Project

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Table of contents

1 Overview **2**

2 Empirical Bayes Filing-Rate Memo **2**

3 Data Construction **3**
3.1 Rent Variables . . . 3
3.2 Filing Rate Variables . . . 3

4 Tracking Descriptives **4**
4.1 Movers vs. Stayers . . . 4

5 Eviction Persistence Regressions **4**
5.1 Setup . . . 4
5.2 Regression Specifications . . . 4
5.3 High-Filer and Rent Regressions . . . 6

6 Inflow and Outflow Analysis **6**
6.1 Leave-One-Out Tract Eviction Rate . . . 6
6.2 Eviction Intensity Bins . . . 7
6.3 Inflow: Where Do Tenants at High-Eviction Buildings Come From? . . . 7
6.4 Outflow: Where Do Tenants From High-Eviction Buildings Go? . . . 7

7 Three-Move Trajectory Analysis **7**
7.1 Design . . . 7
7.2 Descriptive Summary . . . 8
7.3 Trajectory Regressions . . . 8
7.3.1 Filing Outcomes: Delta Specs (Primary) . . . 10
7.3.2 Filing Outcomes: Destination-Level Specs (Primary) . . . 11
7.3.3 Filing Outcomes: Destination-Level Specs (High-Filing Dummy, Primary) . . . 12
7.3.4 Severe Complaints per Unit: Delta Specs (Additional Outcome) . . . 13
7.3.5 Severe Complaints per Unit: Destination-Level Specs (Additional Outcome) . . . 14
7.3.6 Total Violations per Unit: Delta Specs (Additional Outcome) . . . 15
7.3.7 Total Violations per Unit: Destination-Level Specs (Additional Outcome) . . . 16
7.3.8 Severe Violations per Unit: Delta Specs (Additional Outcome) . . . 17

7.3.9 Severe Violations per Unit: Destination-Level Specs (Additional Outcome) . 18
7.3.10 LOO Tract Filing Rate Outcomes (EB, Secondary Main) 19
7.3.11 LOO Tract Filing Rate Outcomes (EB High-Filing Dummy, Secondary Main) 20

8 Interpretation **20**

9 Appendix: Raw Filing Robustness **21**
9.1 Persistence (Raw Long-Run Filing Rate) 21
9.2 High-Filer and Rent (Raw Long-Run Filing Rate) 23
9.3 Trajectory Filing Outcomes (Raw Long-Run Filing Rate) 23
9.4 LOO Tract Filing Rate Outcomes (Raw Appendix Robustness) 26

1 Overview

This document summarizes the address history analysis from `r/address-history-analysis.R`. The script uses the InfoUSA household panel to track tenants across moves and test whether exposure to high-eviction buildings channels households into worse neighborhoods.

Key questions:

1. Do tenants from high-eviction buildings end up in other high-eviction buildings? (Eviction persistence)
2. What are the demographic characteristics of movers vs. stayers?
3. Do tenants who move *into* high-eviction buildings come from different neighborhoods? (Inflow)
4. Where do tenants go *after* leaving high-eviction buildings? (Outflow)
5. Across a 3-move trajectory (origin → middle → destination), does passing through a high-evicting building shift tenants toward higher-eviction neighborhoods? (Trajectory analysis)

Unit of analysis: Household (familyid) × year panel, with building-level (PID) characteristics merged in.

Key data inputs:

- `infousa_cleaned` — household-year panel with addresses
- `infousa_address_xwalk` — links InfoUSA addresses to parcel IDs
- `bldg_panel_blp` — building characteristics, filing rates, rents, occupancy
- `infousa_race_imputed_person`, `infousa_gender_imputed_person` — demographic imputations

2 Empirical Bayes Filing-Rate Memo

Main filing-intensity regressions in this writeup use `filing_rate_eb_pre_covid` as the default building-level eviction measure.

In the production pipeline (`r/make-analytic-sample.R`, section `6b`), this is built as a Gamma-Poisson Empirical Bayes posterior mean:

$$y_{it} \mid \lambda_i \sim \text{Poisson}(E_{it}\lambda_i), \qquad \lambda_i \sim \text{Gamma}(\alpha, \beta),$$

with pooled pre-2019 sufficient statistics:

$$Y_i = \sum_{t \leq 2019} y_{it}, \qquad E_i = \sum_{t \leq 2019} E_{it},$$

and posterior mean:

$$\mathbb{E}[\lambda_i \mid Y_i, E_i] = \frac{\alpha + Y_i}{\beta + E_i}.$$

City and ZIP priors are estimated by method-of-moments; ZIP priors fall back to city prior when support is thin. Raw long-run filing-rate results are kept as appendix robustness tables only.

3 Data Construction

3.1 Rent Variables

The rent variable `med_rent` is the coalesced value of Altos median listing rent and eviction ongoing rent. The nominal log rent `log_med_rent` is used for descriptive purposes.

For regression analysis, we construct a **hedonic-deflated real rent** (`log_med_rent_real`) by estimating a hedonic price index. We estimate a weighted regression of log rent on year, building type, and census tract fixed effects:

$$\log(\text{rent}_{it}) = \alpha_t + \eta_k + \gamma_g + \varepsilon_{it}$$

where α_t are year fixed effects, η_k building type, and γ_g census tract. The year FE $\hat{\alpha}_t$ capture aggregate rent inflation. Real rent is:

$$\log(\text{rent}_{it}^{\text{real}}) = \log(\text{rent}_{it}) - (\hat{\alpha}_t - \hat{\alpha}_{t_0})$$

where t_0 is the earliest year in the sample.

3.2 Filing Rate Variables

Three filing rate measures are available:

- **`filing_rate_raw`**: Annual filings / occupied units. Year-varying, volatile, subject to small-denominator noise.
- **`filing_rate_longrun_pre2019`**: Total pre-2019 filings / (total units × years observed pre-2019). Time-invariant building-level average. Noisier for buildings with short panels.
- **`filing_rate_eb_pre_covid`**: Empirical Bayes shrinkage estimator (Gamma-Poisson model), pre-2019, city-wide prior. Time-invariant. Shrinks noisy short-panel rates toward the city mean.

Main-text filing regressions use `filing_rate_eb_pre_covid`. Raw long-run filing-rate regressions (`filing_rate_longrun_pre2019`) are reported in the appendix for robustness.

4 Tracking Descriptives

4.1 Movers vs. Stayers

The script identifies rental households and classifies them as *movers* (2+ addresses over the panel) or *stayers* (1 address). Summary statistics are computed for each group.

Table 1: Tracking Descriptives: Movers vs. Stayers

	All Rental HHs	Stayers (1 address)	Movers (2+ addresses)
N HH-Years	4,527,988	3,627,712	900,276
N Unique HHs	1,180,429	1,102,507	77,922
Mean Addresses per HH	1.21	1	4.12
Mean Years Observed	3.47	3.29	5.97
% Black (mean posterior)	34.3	34.2	34.9
% White (mean posterior)	57.1	56.9	58.6
% Hispanic (mean posterior)	5.7	5.8	5
% Female HoH	55	54.2	61.5
Mean Filing Rate	0.0692	0.0742	0.0492
Mean Building Size (units)	22.8	22.5	24
Mean Log Rent	7.159	7.139	7.231
N Unique Tracts	418	415	402

5 Eviction Persistence Regressions

5.1 Setup

The core persistence analysis asks: conditional on a household *moving*, does the eviction filing rate at their **previous** building predict the filing rate at their **current** building?

Sample: Rental-to-rental movers (both origin and current address are rental buildings). Filing rates are pre-COVID, time-invariant building characteristics (not annual rates).

5.2 Regression Specifications

The persistence regressions estimate:

$$\text{FilingRate}^{EB}_{i,\text{dest}} = \beta \cdot \text{FilingRate}^{EB}_{i,\text{origin}} + \gamma' X_i + \alpha_{\text{tract,dest}} + \alpha_{\text{tract,orig}} + \varepsilon_i$$

where FilingRate^{EB} is the EB shrinkage filing rate, X_i includes hedonic-deflated real rent at origin and destination, and α_{tract} are census tract fixed effects (used instead of block group to avoid over-absorption with building-level time-invariant rates). Standard errors are clustered by destination PID.

Specifications:

- **m1** (bivariate): Filing rate at destination $\sim$ filing rate at origin
- **m2** (+ FE): Add unit bin and census tract fixed effects (current and previous)
- **m3** (+ real rent): Add hedonic-deflated log rent at origin and destination

Raw long-run versions of these same specs are shown in the appendix.

Table 2: Eviction Persistence: EB Shrinkage Filing Rates (Main)

Dependent Variable:	Filing rate (destination, EB)		
	m1	m2	m3
Model:	(1)	(2)	(3)
Variables			
Filing rate (origin, EB)	0.1063***	0.0486***	0.0442***
	(0.0137)	(0.0086)	(0.0101)
Avg real rent (destination building)			-0.0594***
			(0.0029)
Avg real rent (origin building)			0.0006
			(0.0029)
Fixed-effects			
Units bin (destination)		Yes	Yes
Units bin (origin)		Yes	Yes
Destination tract FE		Yes	Yes
Origin tract FE		Yes	Yes
Fit statistics			
Observations	87,910	87,897	56,361

Clustered (pid) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

5.3 High-Filer and Rent Regressions

The high-filer regressions use a binary indicator ($\geq 15\%$ EB filing rate) in place of the continuous rate:

- **m4**: High-filer (dest) $\sim$ high-filer (origin) + FE
- **m5**: Add hedonic-deflated rent controls, restrict to multi-unit buildings
- **m_rent**: Real rent (dest) $\sim$ filing rate (origin) + real rent (origin) + filing rate (dest) + FE

Table 3: High-Filer Persistence and Rent Regressions (EB Main)

Dependent Variables:	high_filing_dest		Avg real rent (destination building)
	m4	m5	m_rent
Model:	(1)	(2)	(3)
Variables			
high_filing_origTRUE	0.0566***	0.0690***	
	(0.0049)	(0.0114)	
Avg real rent (destination building)		-0.0677***	
		(0.0170)	
Avg real rent (origin building)		-0.0021	0.2186***
		(0.0062)	(0.0071)
Filing rate (origin, EB)			-0.0182**
			(0.0075)
Filing rate (destination, EB)			-0.2619***
			(0.0353)
Fixed-effects			
Units bin (destination)	Yes	Yes	Yes
Units bin (origin)	Yes	Yes	Yes
Destination tract FE	Yes	Yes	Yes
Origin tract FE	Yes	Yes	Yes
Fit statistics			
Observations	87,897	20,186	56,361

Clustered (pid) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

6 Inflow and Outflow Analysis

6.1 Leave-One-Out Tract Eviction Rate

For each building, the LOO tract eviction rate is:

$$\text{LOO}_i = \frac{\sum_{j \in \text{tract}(i), j \neq i} \text{filings}_j}{\sum_{j \in \text{tract}(i), j \neq i} \text{unit-years}_j}$$

This measures the eviction environment of the building's neighborhood, excluding the building's own contribution.

6.2 Eviction Intensity Bins

Buildings are classified using the empirical Bayes eviction filing rate (pre-COVID):

Bin	Definition
[0,1)	EB rate $\in [0, 0.01)$
[1,5)	EB rate $\in [0.01, 0.05)$
[5,10)	EB rate $\in [0.05, 0.10)$
[10,20)	EB rate $\in [0.10, 0.20)$
[20,inf)	EB rate ≥ 0.20

6.3 Inflow: Where Do Tenants at High-Eviction Buildings Come From?

Table 5: Origin characteristics of movers, by destination eviction intensity

Destination bin	N moves	Origin % Black	Dest % Black	Origin LOO evict	Dest LOO evict	Mover % Black	Mover % F
[0,1)	51,525	0.327	0.331	0.0390	0.0388	0.247	
[1,5)	8,215	0.280	0.276	0.0356	0.0377	0.176	
[5,10)	12,252	0.480	0.497	0.0526	0.0549	0.448	
[10,20)	10,109	0.505	0.523	0.0563	0.0587	0.480	
[20,inf)	5,879	0.578	0.569	0.0643	0.0732	0.547	

6.4 Outflow: Where Do Tenants From High-Eviction Buildings Go?

Table 6: Destination characteristics of movers, by origin eviction intensity

Origin bin	N moves	Origin LOO evict	Dest LOO evict	Δ LOO tract	Mover % Black	Mover % Female
[0,1)	49,277	0.0376	0.0405	0.0029	0.255	0.583
[1,5)	8,759	0.0337	0.0333	-0.0004	0.157	0.572
[5,10)	13,103	0.0519	0.0533	0.0014	0.416	0.636
[10,20)	10,813	0.0573	0.0568	-0.0005	0.468	0.646
[20,inf)	6,191	0.0725	0.0658	-0.0067	0.547	0.664

7 Three-Move Trajectory Analysis

7.1 Design

The trajectory analysis tracks households across three consecutive residential spells:

$$\text{Origin} \xrightarrow{\text{move 1}} \text{Middle} \xrightarrow{\text{move 2}} \text{Destination}$$

The key question: conditional on origin neighborhood, does the eviction intensity of the **middle** building predict worse outcomes at the **destination** building?

Primary filing outcomes use building-level EB filing rates directly, capturing within-neighborhood sorting:

- $\Delta\text{FR}^{\text{EB}} = \text{FR}^{\text{EB}}_{\text{dest}} - \text{FR}^{\text{EB}}_{\text{origin}}$ (change in EB filing rate)

Raw long-run filing-rate versions are reported in the appendix as robustness checks.

Secondary outcome uses the LOO tract eviction rate (neighborhood-level, excluding the focal building):

$$\Delta\text{LOO} = \text{LOO}_{\text{dest}} - \text{LOO}_{\text{origin}}$$

The LOO rate averages over all buildings in the tract and thus misses within-tract sorting — a tenant moving between two high-eviction buildings in the same tract shows zero change in LOO. Building-level rates capture this directly.

7.2 Descriptive Summary

Table 7: Three-move trajectories by middle building eviction intensity

Middle bin	N traj.	Orig EB	Dest EB	Δ EB	Orig LOO (EB)	Dest LOO (EB)	Δ LOO (EB)	Orig LO
[0,1)	104,812	0.0381	0.0398	0.0017	0.0491	0.0492	2e-04	
[1,5)	9,196	0.0417	0.0406	-0.0011	0.0451	0.0453	2e-04	
[5,10)	23,030	0.0544	0.0552	0.0008	0.0578	0.0582	4e-04	
[10,20)	16,075	0.0643	0.0672	0.0029	0.0604	0.0609	5e-04	
[20,inf)	7,037	0.0930	0.0991	0.0061	0.0652	0.0656	4e-04	

7.3 Trajectory Regressions

For each outcome family, the script estimates delta and destination-level models separately, with progressively richer origin conditioning. Filing-outcome models are estimated using both:

- middle-building filing bins (`[0,1)`, `[1,5)`, `[5,10)`, `[10,20)`, `[20,inf)`)
- a high-filing dummy (`\ge 15\%`)

Delta specs:

(A) Unconditional delta:

$$\Delta Y_i = \sum_b \beta_b \cdot \mathbf{1}[\text{mid_bin}_i = b] + \varepsilon_i$$

(B) Delta + origin tract FE:

$$\Delta Y_i = \sum_b \beta_b \cdot \mathbf{1}[\text{mid_bin}_i = b] + \alpha_{\text{tract,orig}} + \varepsilon_i$$

(C) Delta + origin filing rate + origin parcel controls:

$$\Delta Y_i = \sum_b \beta_b \cdot \mathbf{1}[\text{mid_bin}_i = b] + \theta Y_{\text{orig},i} + \gamma' X_{\text{orig},i} + \alpha_{\text{tract,orig}} + \varepsilon_i$$

(D) Delta + origin-tract-by-origin-filing-bin FE:

$$\Delta Y_i = \sum_b \beta_b \cdot \mathbf{1}[\text{mid_bin}_i = b] + \gamma' X_{\text{orig},i} + \alpha_{\text{tract,orig}\times\text{orig_filing_bin}} + \varepsilon_i$$

Destination-level specs:

(A) Destination level + origin tract FE:

$$Y_{\text{dest},i} = \sum_b \beta_b \cdot \mathbf{1}[\text{mid_bin}_i = b] + \alpha_{\text{tract,orig}} + \varepsilon_i$$

(B) + origin outcome level:

$$Y_{\text{dest},i} = \sum_b \beta_b \cdot \mathbf{1}[\text{mid_bin}_i = b] + \theta Y_{\text{orig},i} + \alpha_{\text{tract,orig}} + \varepsilon_i$$

(C) + origin parcel controls:

$$Y_{\text{dest},i} = \sum_b \beta_b \cdot \mathbf{1}[\text{mid_bin}_i = b] + \theta Y_{\text{orig},i} + \gamma' X_{\text{orig},i} + \alpha_{\text{tract,orig}} + \varepsilon_i$$

(D) + origin-tract-by-origin-filing-bin FE:

$$Y_{\text{dest},i} = \sum_b \beta_b \cdot \mathbf{1}[\text{mid_bin}_i = b] + \gamma' X_{\text{orig},i} + \alpha_{\text{tract,orig}\times\text{orig_filing_bin}} + \varepsilon_i$$

where Y is one of: building-level filing rate, severe complaints per unit, total violations per unit, severe violations per unit, or LOO tract filing rate (EB or raw). Origin parcel controls $X_{\text{orig},i}$ include `log(total_area)`, `log(market_value)`, `year_blt_decade`, `num_units_bin`, `building_type`, `quality_grade_standard`, and `num_stories_bin`. Reference category for bin models: [0,1). Standard errors are clustered by origin block group.

7.3.1 Filing Outcomes: Delta Specs (Primary)

Dependent Variable:	Change in filing rate (destination - origin, EB)			
	Delta A	Delta B	Delta C	Delta D
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	0.0017***			
	(0.0003)			
Middle building filing-rate bin = [1,5)")	-0.0028*	-0.0030*	0.0017	0.0022*
	(0.0017)	(0.0017)	(0.0011)	(0.0012)
Middle building filing-rate bin = [5,10)")	-0.0009	-0.0004	0.0048***	0.0043***
	(0.0010)	(0.0010)	(0.0007)	(0.0008)
Middle building filing-rate bin = [10,20)")	0.0012	0.0018	0.0135***	0.0121***
	(0.0013)	(0.0013)	(0.0010)	(0.0012)
Middle building filing-rate bin = [20,inf)")	0.0044*	0.0053**	0.0364***	0.0297***
	(0.0026)	(0.0026)	(0.0022)	(0.0025)
Origin filing rate (EB)			-0.9401***	
			(0.0079)	
Log(origin total area)			0.0030***	-0.0003
			(0.0007)	(0.0009)
Log(origin market value)			-0.0071***	-0.0098***
			(0.0015)	(0.0019)
Fixed-effects				
Origin tract FE		Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,137	160,132	157,330	157,262

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.2 Filing Outcomes: Destination-Level Specs (Primary)

Dependent Variable:	Destination filing rate (EB)			
	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)
Variables				
Middle building filing-rate bin = [1,5)")	0.0055***	0.0049***	0.0017	0.0024**
	(0.0011)	(0.0011)	(0.0011)	(0.0011)
Middle building filing-rate bin = [5,10)")	0.0068***	0.0063***	0.0048***	0.0048***
	(0.0009)	(0.0008)	(0.0007)	(0.0008)
Middle building filing-rate bin = [10,20)")	0.0163***	0.0153***	0.0135***	0.0129***
	(0.0011)	(0.0010)	(0.0010)	(0.0010)
Middle building filing-rate bin = [20,inf)")	0.0442***	0.0416***	0.0364***	0.0341***
	(0.0026)	(0.0024)	(0.0022)	(0.0021)
Origin filing rate (EB)		0.0688***	0.0599***	
		(0.0089)	(0.0079)	
Log(origin total area)			0.0030***	0.0026***
			(0.0007)	(0.0007)
Log(origin market value)			-0.0071***	-0.0063***
			(0.0015)	(0.0014)
Fixed-effects				
Origin tract FE	Yes	Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,132	160,132	157,330	157,262

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.3 Filing Outcomes: Destination-Level Specs (High-Filing Dummy, Primary)

Dependent Variable:	Destination filing rate (EB)			
	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)
Variables				
Middle building high-filing indicator (>=15%)	0.0312***	0.0292***	0.0258***	0.0242***
	(0.0018)	(0.0017)	(0.0015)	(0.0014)
Origin filing rate (EB)		0.0711***	0.0615***	
		(0.0091)	(0.0080)	
Log(origin total area)			0.0030***	0.0027***
			(0.0007)	(0.0007)
Log(origin market value)			-0.0073***	-0.0065***
			(0.0015)	(0.0015)
Fixed-effects				
Origin tract FE	Yes	Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,132	160,132	157,330	157,262

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.4 Severe Complaints per Unit: Delta Specs (Additional Outcome)

Dependent Variable:	Change in severe complaints per unit			
	Delta A	Delta B	Delta C	Delta D
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	8.02×10^{-5}			
	(0.0034)			
Middle building filing-rate bin = [1,5)")	-0.0041	-0.0078	-0.0105	-0.0272
	(0.0162)	(0.0166)	(0.0121)	(0.0176)
Middle building filing-rate bin = [5,10)")	-0.0169	-0.0128	0.0158*	0.0047
	(0.0116)	(0.0120)	(0.0092)	(0.0121)
Middle building filing-rate bin = [10,20)")	-0.0132	-0.0080	0.0492***	0.0247*
	(0.0139)	(0.0142)	(0.0110)	(0.0143)
Middle building filing-rate bin = [20,inf)")	-0.0433*	-0.0382*	0.0558***	0.0302
	(0.0225)	(0.0228)	(0.0166)	(0.0237)
Origin severe complaints per unit			-0.9584***	
			(0.0040)	
Log(origin total area)			0.0144*	-0.0504***
			(0.0082)	(0.0117)
Log(origin market value)			0.0043	-0.0196*
			(0.0094)	(0.0115)
Fixed-effects				
Origin tract FE		Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,284	160,279	157,468	157,338

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.5 Severe Complaints per Unit: Destination-Level Specs (Additional Outcome)

Dependent Variable:	Destination severe complaints per unit			
	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)
Variables				
Middle building filing-rate bin = [1,5)")	-0.0154	-0.0150	-0.0105	-0.0054
	(0.0119)	(0.0119)	(0.0121)	(0.0122)
Middle building filing-rate bin = [5,10)")	0.0197**	0.0182**	0.0158*	0.0148
	(0.0091)	(0.0091)	(0.0092)	(0.0093)
Middle building filing-rate bin = [10,20)")	0.0527***	0.0500***	0.0492***	0.0494***
	(0.0110)	(0.0109)	(0.0110)	(0.0111)
Middle building filing-rate bin = [20,inf)")	0.0648***	0.0603***	0.0558***	0.0553***
	(0.0168)	(0.0167)	(0.0166)	(0.0167)
Origin severe complaints per unit		0.0439***	0.0416***	
		(0.0039)	(0.0040)	
Log(origin total area)			0.0144*	0.0175**
			(0.0082)	(0.0084)
Log(origin market value)			0.0043	0.0044
			(0.0094)	(0.0098)
Fixed-effects				
Origin tract FE	Yes	Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,279	160,279	157,468	157,338

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.6 Total Violations per Unit: Delta Specs (Additional Outcome)

Dependent Variable:	Change in total violations per unit			
	Delta A	Delta B	Delta C	Delta D
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	0.0104			
	(0.0154)			
Middle building filing-rate bin = [1,5)")	-0.0237	-0.0366	-0.0505	-0.1696**
	(0.0674)	(0.0688)	(0.0538)	(0.0747)
Middle building filing-rate bin = [5,10)")	-0.0787	-0.0654	0.0673	0.0045
	(0.0546)	(0.0559)	(0.0413)	(0.0562)
Middle building filing-rate bin = [10,20)")	-0.0100	0.0101	0.2734***	0.1270*
	(0.0666)	(0.0677)	(0.0530)	(0.0680)
Middle building filing-rate bin = [20,inf)")	-0.0644	-0.0479	0.2546***	0.1695*
	(0.0935)	(0.0950)	(0.0768)	(0.0987)
Origin total violations per unit			-0.9445***	
			(0.0040)	
Log(origin total area)			-0.0871**	-0.3360***
			(0.0342)	(0.0520)
Log(origin market value)			0.0685	0.1933***
			(0.0421)	(0.0584)
Fixed-effects				
Origin tract FE		Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,284	160,279	157,468	157,338

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.7 Total Violations per Unit: Destination-Level Specs (Additional Outcome)

Dependent Variable:	Destination total violations per unit			
	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)
Variables				
Middle building filing-rate bin = [1,5)")	-0.0809	-0.0783	-0.0505	-0.0388
	(0.0530)	(0.0525)	(0.0538)	(0.0541)
Middle building filing-rate bin = [5,10)")	0.0873**	0.0781*	0.0673	0.0680
	(0.0417)	(0.0415)	(0.0413)	(0.0416)
Middle building filing-rate bin = [10,20)")	0.2879***	0.2711***	0.2734***	0.2785***
	(0.0532)	(0.0527)	(0.0530)	(0.0540)
Middle building filing-rate bin = [20,inf)")	0.2606***	0.2420***	0.2546***	0.2513***
	(0.0771)	(0.0761)	(0.0768)	(0.0777)
Origin total violations per unit		0.0604***	0.0555***	
		(0.0039)	(0.0040)	
Log(origin total area)			-0.0871**	-0.0676*
			(0.0342)	(0.0351)
Log(origin market value)			0.0685	0.0507
			(0.0421)	(0.0443)
Fixed-effects				
Origin tract FE	Yes	Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,279	160,279	157,468	157,338

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.8 Severe Violations per Unit: Delta Specs (Additional Outcome)

Dependent Variable:	Change in severe violations per unit			
	Delta A	Delta B	Delta C	Delta D
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	-0.0013			
	(0.0048)			
Middle building filing-rate bin = [1,5)")	0.0014	-0.0039	0.0049	-0.0159
	(0.0190)	(0.0195)	(0.0163)	(0.0213)
Middle building filing-rate bin = [5,10)")	-0.0188	-0.0121	0.0134	0.0009
	(0.0173)	(0.0176)	(0.0118)	(0.0177)
Middle building filing-rate bin = [10,20)")	0.0066	0.0132	0.0401***	0.0373*
	(0.0189)	(0.0195)	(0.0144)	(0.0198)
Middle building filing-rate bin = [20,inf)")	0.0209	0.0289	0.0668***	0.0787**
	(0.0306)	(0.0312)	(0.0253)	(0.0321)
Origin severe violations per unit			-0.9867***	
			(0.0038)	
Log(origin total area)			-0.0100	-0.0270**
			(0.0089)	(0.0119)
Log(origin market value)			0.0164	0.0042
			(0.0118)	(0.0168)
Fixed-effects				
Origin tract FE		Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,284	160,279	157,468	157,338

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.9 Severe Violations per Unit: Destination-Level Specs (Additional Outcome)

Dependent Variable:	Destination severe violations per unit			
	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)
Variables				
Middle building filing-rate bin = [1,5)")	0.0112	0.0110	0.0049	0.0072
	(0.0155)	(0.0155)	(0.0163)	(0.0165)
Middle building filing-rate bin = [5,10)")	0.0162	0.0157	0.0134	0.0132
	(0.0117)	(0.0117)	(0.0118)	(0.0118)
Middle building filing-rate bin = [10,20)")	0.0433***	0.0428***	0.0401***	0.0406***
	(0.0143)	(0.0143)	(0.0144)	(0.0146)
Middle building filing-rate bin = [20,inf)")	0.0728***	0.0722***	0.0668***	0.0662***
	(0.0253)	(0.0252)	(0.0253)	(0.0256)
Origin severe violations per unit		0.0152***	0.0133***	
		(0.0038)	(0.0038)	
Log(origin total area)			-0.0100	-0.0099
			(0.0089)	(0.0091)
Log(origin market value)			0.0164	0.0148
			(0.0118)	(0.0124)
Fixed-effects				
Origin tract FE	Yes	Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,279	160,279	157,468	157,338

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.10 LOO Tract Filing Rate Outcomes (EB, Secondary Main)

Dependent Variables:	Change in LOO tract filing rate (EB)		Destination LOO tract filing rate (EB)			
	Delta A	Delta B	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Constant	0.0002					
	(0.0002)					
Middle building filing-rate bin = [1,5)")	-1.31×10^{-5}	0.0002	0.0002	0.0002	0.0002	0.0002
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0002)
Middle building filing-rate bin = [5,10)")	0.0002	0.0008***	0.0007***	0.0007***	0.0007***	0.0007***
	(0.0002)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Middle building filing-rate bin = [10,20)")	0.0003	0.0011***	0.0010***	0.0010***	0.0010***	0.0010***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Middle building filing-rate bin = [20,inf)")	0.0002	0.0016***	0.0011***	0.0013***	0.0013***	0.0014***
	(0.0004)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Origin LOO tract filing rate (EB)				0.3713**	0.3631**	
				(0.1508)	(0.1619)	
Log(origin total area)					3.81×10^{-5}	4.12×10^{-5}
					(0.0002)	(0.0002)
Log(origin market value)					-0.0007***	-0.0007***
					(0.0002)	(0.0002)
Fixed-effects						
Origin tract FE		Yes	Yes	Yes	Yes	
Origin year-built decade FE					Yes	Yes
Origin unit-bin FE					Yes	Yes
Origin building-type FE					Yes	Yes
Origin quality-grade FE					Yes	Yes
Origin stories-bin FE					Yes	Yes
Origin tract x origin filing-bin FE						Yes
Fit statistics						
Observations	160,667	160,658	160,658	160,658	157,691	157,563

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.3.11 LOO Tract Filing Rate Outcomes (EB High-Filing Dummy, Secondary Main)

Dependent Variables:	Change in LOO tract filing rate (EB)		Destination LOO tract filing rate (EB)			
	Delta A	Delta B	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Constant	0.0002					
	(0.0002)					
Middle building high-filing indicator (>=15%)	0.0002	0.0012***	0.0009***	0.0010***	0.0010***	0.0011***
	(0.0003)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Origin LOO tract filing rate (EB)				0.3694**	0.3615**	
				(0.1508)	(0.1619)	
Log(origin total area)					4.22×10^{-5}	4.34×10^{-5}
					(0.0002)	(0.0002)
Log(origin market value)					-0.0007***	-0.0008***
					(0.0002)	(0.0002)
Fixed-effects						
Origin tract FE		Yes	Yes	Yes	Yes	
Origin year-built decade FE					Yes	Yes
Origin unit-bin FE					Yes	Yes
Origin building-type FE					Yes	Yes
Origin quality-grade FE					Yes	Yes
Origin stories-bin FE					Yes	Yes
Origin tract x origin filing-bin FE						Yes
Fit statistics						
Observations	160,667	160,658	160,658	160,658	157,691	157,563

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

8 Interpretation

Eviction persistence: The positive coefficient on previous filing rate indicates that tenants who leave high-eviction buildings tend to end up in other high-eviction buildings, even after controlling for tract and building size. This is not simply a neighborhood effect, and persists within origin-tract conditioning. EB is the main measure shown; raw long-run robustness is in the appendix.

Hedonic rent deflation: Using hedonic-deflated real rents (rather than nominal) ensures that rent persistence results are not confounded by aggregate rent inflation over the panel period. The year fixed effects from the hedonic regression capture common rent trends, so `log_med_rent_real` isolates cross-sectional and within-building rent variation.

Inflow/Outflow: High-eviction buildings draw tenants from neighborhoods that are already higher-eviction, and their departing tenants move to similar or worse neighborhoods. The LOO tract eviction rate captures the neighborhood environment excluding the focal building.

Trajectories: The trajectory design reports delta and destination-level outcomes separately, and adds richer origin conditioning (origin filing rate, origin parcel controls, and origin-tract-by-origin-filing-bin FE). This clarifies whether sorting patterns persist after conditioning on baseline origin-building risk and building characteristics. We also add severe complaints per unit as an additional long-run building-level outcome. Unlike the LOO measure, building-level outcomes capture within-neighborhood sorting, so tenants may systematically sort into worse buildings even within the same tract.

9 Appendix: Raw Filing Robustness

9.1 Persistence (Raw Long-Run Filing Rate)

Table 8: Eviction Persistence: Raw Long-Run Filing Rates (Appendix Robustness)

Dependent Variable:	Filing rate (destination, raw long-run)		
	m1	m2	m3
Model:	(1)	(2)	(3)
Variables			
Filing rate (origin, raw long-run)	0.1069***	0.0466***	0.0447***
	(0.0162)	(0.0092)	(0.0114)
Avg real rent (destination building)			-0.0599***
			(0.0028)
Avg real rent (origin building)			0.0016
			(0.0029)
Fixed-effects			
Units bin (destination)		Yes	Yes
Units bin (origin)		Yes	Yes
Destination tract FE		Yes	Yes
Origin tract FE		Yes	Yes
Fit statistics			
Observations	87,989	87,976	56,412

Clustered (pid) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 9: High-Filer Persistence and Rent Regressions (Raw Long-Run Appendix Robustness)

Dependent Variables:	high_filing_dest		Avg real rent (destination building)
	m4	m5	m_rent
Model:	(1)	(2)	(3)
Variables			
high_filing_origTRUE	0.0571***	0.0698***	
	(0.0052)	(0.0140)	
Avg real rent (destination building)		-0.0180***	
		(0.0063)	
Avg real rent (origin building)		-0.0033	0.2192***
		(0.0044)	(0.0072)
Filing rate (origin, raw long-run)			-0.0130*
			(0.0068)
Filing rate (destination, raw long-run)			-0.2731***
			(0.0399)
Fixed-effects			
Units bin (destination)	Yes	Yes	Yes
Units bin (origin)	Yes	Yes	Yes
Destination tract FE	Yes	Yes	Yes
Origin tract FE	Yes	Yes	Yes
Fit statistics			
Observations	87,976	20,209	56,412

Clustered (pid) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

9.2 High-Filer and Rent (Raw Long-Run Filing Rate)

9.3 Trajectory Filing Outcomes (Raw Long-Run Filing Rate)

Dependent Variable:	Change in filing rate (destination - origin, raw long-run)			
	Delta A	Delta B	Delta C	Delta D
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	0.0027***			
	(0.0005)			
Middle building filing-rate bin = [1,5)")	-0.0039	-0.0053	0.0006	0.0018
	(0.0057)	(0.0052)	(0.0028)	(0.0031)
Middle building filing-rate bin = [5,10)")	-0.0005	0.0009	0.0080***	0.0067***
	(0.0017)	(0.0019)	(0.0017)	(0.0015)
Middle building filing-rate bin = [10,20)")	0.0008	0.0025	0.0164***	0.0145***
	(0.0017)	(0.0018)	(0.0021)	(0.0021)
Middle building filing-rate bin = [20,inf)")	0.0027	0.0048*	0.0398***	0.0309***
	(0.0029)	(0.0028)	(0.0044)	(0.0031)
Origin filing rate (raw long-run)			-0.9800***	
			(0.0158)	
Log(origin total area)			0.0025*	-0.0011
			(0.0013)	(0.0020)
Log(origin market value)			-0.0040	-0.0070***
			(0.0026)	(0.0026)
Fixed-effects				
Origin tract FE		Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,284	160,279	157,468	157,338

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Dependent Variable:	Destination filing rate (raw long-run)			
	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)
Variables				
Middle building filing-rate bin = [1,5)")	0.0039	0.0037	0.0006	0.0022
	(0.0025)	(0.0025)	(0.0028)	(0.0029)
Middle building filing-rate bin = [5,10)")	0.0098***	0.0096***	0.0080***	0.0076***
	(0.0018)	(0.0019)	(0.0017)	(0.0015)
Middle building filing-rate bin = [10,20)")	0.0188***	0.0184***	0.0164***	0.0153***
	(0.0022)	(0.0024)	(0.0021)	(0.0019)
Middle building filing-rate bin = [20,inf)")	0.0463***	0.0452***	0.0398***	0.0365***
	(0.0049)	(0.0054)	(0.0044)	(0.0038)
Origin filing rate (raw long-run)		0.0250	0.0200	
		(0.0177)	(0.0158)	
Log(origin total area)			0.0025*	0.0015
			(0.0013)	(0.0012)
Log(origin market value)			-0.0040	-0.0038*
			(0.0026)	(0.0022)
Fixed-effects				
Origin tract FE	Yes	Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,279	160,279	157,468	157,338

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Dependent Variable:	Destination filing rate (raw long-run)			
	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)
Variables				
Middle building high-filing indicator (>=15%)	0.0349***	0.0341***	0.0305***	0.0277***
	(0.0038)	(0.0042)	(0.0035)	(0.0030)
Origin filing rate (raw long-run)		0.0262	0.0208	
		(0.0180)	(0.0161)	
Log(origin total area)			0.0026*	0.0016
			(0.0013)	(0.0012)
Log(origin market value)			-0.0043	-0.0040*
			(0.0026)	(0.0022)
Fixed-effects				
Origin tract FE	Yes	Yes	Yes	
Origin year-built decade FE			Yes	Yes
Origin unit-bin FE			Yes	Yes
Origin building-type FE			Yes	Yes
Origin quality-grade FE			Yes	Yes
Origin stories-bin FE			Yes	Yes
Origin tract x origin filing-bin FE				Yes
Fit statistics				
Observations	160,279	160,279	157,468	157,338

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

9.4 LOO Tract Filing Rate Outcomes (Raw Appendix Robustness)

Dependent Variables:	Change in LOO tract filing rate (raw)		Destination LOO tract filing rate (raw)			
	Delta A	Delta B	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Constant	-5.33×10^{-5}					
	(0.0006)					
Middle building filing-rate bin = [1,5)")	-0.0004	0.0002	2.54×10^{-5}	0.0001	0.0003	0.0004
	(0.0016)	(0.0008)	(0.0009)	(0.0008)	(0.0005)	(0.0005)
Middle building filing-rate bin = [5,10)")	-3.01×10^{-5}	0.0003	0.0002	0.0003	0.0004	0.0004
	(0.0002)	(0.0003)	(0.0003)	(0.0003)	(0.0002)	(0.0002)
Middle building filing-rate bin = [10,20)")	2.42×10^{-5}	0.0006	0.0005	0.0006	0.0006	0.0006*
	(0.0003)	(0.0004)	(0.0005)	(0.0004)	(0.0004)	(0.0004)
Middle building filing-rate bin = [20,inf)")	-9.52×10^{-5}	0.0011***	0.0006	0.0009**	0.0011***	0.0009***
	(0.0004)	(0.0004)	(0.0005)	(0.0004)	(0.0003)	(0.0003)
Origin LOO tract filing rate (raw)				0.7790***	0.8058***	
				(0.1025)	(0.1230)	
Log(origin total area)					-0.0004	-0.0005
					(0.0005)	(0.0004)
Log(origin market value)					-0.0004	-0.0004
					(0.0004)	(0.0003)
Fixed-effects						
Origin tract FE		Yes	Yes	Yes	Yes	
Origin year-built decade FE					Yes	Yes
Origin unit-bin FE					Yes	Yes
Origin building-type FE					Yes	Yes
Origin quality-grade FE					Yes	Yes
Origin stories-bin FE					Yes	Yes
Origin tract x origin filing-bin FE						Yes
Fit statistics						
Observations	160,667	160,658	160,658	160,658	157,691	157,563

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Dependent Variables:	Change in LOO tract filing rate (raw)		Destination LOO tract filing rate (raw)			
	Delta A	Delta B	Dest A	Dest B	Dest C	Dest D
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Constant	-6.88×10^{-5}					
	(0.0006)					
Middle building high-filing indicator (>=15%)	-0.0002	0.0006	0.0003	0.0006	0.0007**	0.0006*
	(0.0003)	(0.0004)	(0.0005)	(0.0004)	(0.0003)	(0.0003)
Origin LOO tract filing rate (raw)				0.7780***	0.8049***	
				(0.1033)	(0.1233)	
Log(origin total area)					-0.0004	-0.0005
					(0.0005)	(0.0004)
Log(origin market value)					-0.0004	-0.0004
					(0.0004)	(0.0003)
Fixed-effects						
Origin tract FE		Yes	Yes	Yes	Yes	
Origin year-built decade FE					Yes	Yes
Origin unit-bin FE					Yes	Yes
Origin building-type FE					Yes	Yes
Origin quality-grade FE					Yes	Yes
Origin stories-bin FE					Yes	Yes
Origin tract x origin filing-bin FE						Yes
Fit statistics						
Observations	160,667	160,658	160,658	160,658	157,691	157,563

Clustered (Origin block group) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*